



Course Number : **EEE G626**
Course Title : **Hardware and Software Codesign**
Instructor-In-Charge : **Prof. Vinay Chamola**

Note: This course is a highly specialized master's degree course (**Level 6 course**) and assumes that the students taking up the course have solid background of FPGA/ embedded system design etc. Undergraduate students willing to take this course are strongly recommended to contact the IC and take prior permission in case they are interested to take this course (vinay.chamola@pilani.bits-pilani.ac.in) so as to avoid trouble for them later on.

1. Scope and Objective:

This course deals with the system-level design of embedded systems comprised of both hardware and software. It investigates topics ranging from system modeling to hardware-software implementation. This course also teaches analysis and optimization processes for supporting algorithmic and architectural design decisions. It also helps the student gain design experience through different case studies. The course emphasizes a top-down design methodology driven by bottom-up constraints. The topics covered will include co-design of hardware and software involving key aspects like partitioning, mapping and optimization.

2. Course Learning Objectives (CLOs) are as follows

- Analyze embedded system requirements and develop suitable hardware–software co-design solutions by applying system-level modeling, partitioning, and synthesis techniques.
- Evaluate and optimize hardware–software architectures using partitioning algorithms, mapping strategies, and performance estimation methods to satisfy design constraints such as speed, area, power, and cost.
- Design and implement hardware–software co-designed embedded applications using contemporary design methodologies and FPGA-based platforms to address real-world engineering problems.

2. Text Book:

- (T1) Daniel D Gajski, Frank Vahid, Sanjay Narayan, Jie Gong, *Specification and Design of Embedded Systems*, Prentice Hall, 1994.
(T2) Peter Marwedel, *Embedded System Design*, Kluwer Academic Publishers, 2003, ISBN: 1402076908
R1. G. DeMicheli, R. Ernst and W. Wolf, Readings in Hw/Sw Co-design, M. Kaufmann, 2002
R2. Ahmed A. Jerraya and Jean Mermet eds.: System Level Synthesis, Kluwer 1999.
R3. Hardware/Software Codesign. G. DeMicheli and M. Sami (eds.), NATO ASI Series E, 1996.
R4. Sanjaya Kumar, James H. Aylor, Barry W. Johnson, and Wm. A. Wulf. The Codesign of Embedded Systems. Kluwer, 1995





4. Course Plan

Lecture no.	Topics covered
1,2	Introduction to Hardware Software codesign (HSC)
3,4,5,6	Case study to understand Hardware Software codesign
7	Unified HW/SW representation and partitioning
8,9	Hardware Software partitioning
10,11	Synthesis
12,13	Case study: TAM using HSC
14,15,16	Mapping applications to architectures
17,18,19	Partitioning Algorithms -1
20,21,22	Partitioning issues
23,24,25	GCLP algorithm for partitioning
26,27,28	Partitioning methods: ILP, clustering
29, 30, 31, 32	KL Algorithm for partitioning, Simulated annealing etc.
33, 34, 35	Multi criteria optimization
36, 37	Performance estimation
38-40	Other issues in HSC

Labs: There will be interactive labs which will make use of HSC concepts. Various FPGA boards will be used for the same.





Evaluation Scheme:

No	Evaluation Component & Type	Duration	Weightage	Date, Time, Venue	Remark
1.	Midsem	1.5 hours	30%	TBA	CB
3	Lab test	1 hour	15%	TBA	OB
4	Project		15%	TBA	OB
5	Comprehensive Examination	3 hours	40%	TBA	CB +OB

- **Note:** There may be minor changes in the evaluation components/ their weightage based on the progress of the semester/ AGSRD guidelines.

Chamber Consultation Hour: To be announced in Class

Notices: All notices regarding the course will be mailed to the students.

Make-up Policy: No make-up without prior permission.

Instructor-in-charge

